Minkyung Kwon

minkyung.kwon@kaist.ac.kr

Website | LinkedIn | Google Scholar | +82-10-7190-7189

Research Interest

My research interests span generative modeling and 3D representation learning, with a focus on enforcing geometric understanding in diffusion models, 3D mesh representation and generation, and video generation models — particularly world models.

Education

KAIST (Korea Advanced Institute of Science and Technology) 2025.03 – 2027.02 (Expected)
M.S. student, Graduate School of AI (GPA: 4.0 / 4.0) *Seoul, South Korea*
PI: Prof. Seungryong Kim

POSTECH (Pohang University of Science and Technology) 2021.02 – 2025.01
B.S. at Computer Science and Engineering *Pohang, South Korea*
Graduated Cum Laude

Xi'an Hanova International School 2016.04 – 2020.05
International Baccalaureate, Bilingual Diploma *Xi'an, China*
Head Girl

Publications

Conference Papers

[C1] "Seoul World Model: Grounding World Simulation Models in a Real-World Metropolis." ***ECCV 2026 Spotlight Oral***.
Junyoung Seo*, Hyunwook Choi*, **Minkyung Kwon**, Jinhyeok Choi, Siyoon Jin, Gayoung Lee, Junho Kim, JoungBin Lee, Geonmo Gu, Dongyoon Han, Sangdoo Yun, Seungryong Kim†, Jin-Hwa Kim†.

[C2] "Geometric Latent Diffusion: Repurposing Geometric Foundation Models for Multi-view Diffusion." ***ECCV 2026 Long Oral***.
Wooseok Jang, Seonghu Jeon, Jisang Han, Jinhyeok Choi, **Minkyung Kwon**, Seungryong Kim, Saining Xie, Sainan Liu.

[C3] "CAMEO: Correspondence-Attention Alignment for Multi-View Diffusion Models." ***CVPR 2026***.
Minkyung Kwon*, Jinhyeok Choi*, Jiho Park*, Seonghu Jeon, Jinhyuk Jang, Min-Seop Kwak, Junyoung Seo, Seungryong Kim†.

Preprints / Under Review

- "MORPHOS: Autoregressive 4D Generation with Temporal Structured Latents." *arXiv preprint*, 2026.
  Minkyung Kwon*, Jinhyeok Choi*, Youngjin Shin, Jaeyeong Kim, JongMin Lee, Seungryong Kim†.
- "ReNoV: Feature Warping-and-Conditioning for Representation-Guided Novel View Synthesis." *arXiv preprint*, 2026.
  Min-Seop Kwak*, **Minkyung Kwon***, Jinhyeok Choi*, Jiho Park, Seungryong Kim†.

Work Experience

NAVER Corporation 2025.05 - Present
Research Intern, Generation Research Team, Working with Dr. Jin-Hwa Kim

CVLAB, KAIST 2024.08 - 2025.02
Research Intern, Supervised by Prof. Seungryong Kim

Lee Optimization Group, POSTECH 2024.01 - 2024.07
Research Intern, Supervised by Prof. Namhoon Lee

Innowireless 2023.06 – 2023.08
Summer Research Intern. Explored acceleration and compression techniques for Neural Radiance Fields (NeRF) to enable real-time applications. Implemented and evaluated TensoRF for lightweight 3D scene representation, and benchmarked speed–accuracy trade-offs on multi-view datasets.

International Student and Scholar Office, POSTECH 2022 – 2023
Assisted international students in adapting to life in Pohang by providing guidance and support. Planned and organized cultural festivals, local trips, and sports events to promote community engagement.

Honours and Awards

- **Jigok Scholarship**, POSTECH 2021–2025
  Full tuition support, 22,320,000 KRW ($\approx \$14,500$ USD)
- **Best Paper Award**, Department of Humanities and Social Sciences, POSTECH Aug. 2023
  Awarded for an outstanding course paper in the History of Science lecture
- **CES 2024 Travel Award**, POSTECH Jan. 2024
  Travel grant of 3,000,000 KRW ($\approx \$1,900$ USD)

Skills

Languages: Korean (native), English (fluent), Chinese (elementary)

Technical Skills:

- Programming: Python, C, C++
- Frameworks: PyTorch, Unreal Engine, Blender